OdeD at GroundLM 2026 Shared Tasks: Reading the Scorer — Metric Decomposition and Annotation-Convention Recovery for Literature-Grounded QA

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Abstract

We describe team **OdeD**'s submission to LitTraceQA, the literature-grounded question answering task at GroundLM 2026. The task scores three coupled outputs — paper identifiers, coarse evidence locators, and answers in multiple-choice or table form — and our best official score on the 71-question held-out test split is **0.7649**, against **0.4563** for our first submission and **0.5519** for our best fully automated one. Almost none of the improvement came from the retrieval-and-reading pipeline we began with. It came from two other places. First, we read the released evaluator and found the metric is far more asymmetric than it looks: paper F1 and evidence F1 carry weight $1/3$ each while each of the three answer metrics carries only $1/9$, and the evidence key is a coarse tuple in which a correct value on a correct page still scores zero if the *source type* or the visible object id is wrong. Second, we treated the 55-example development split as a corpus for recovering the dataset's *annotation conventions* rather than as a validation set for tuning, and we treated our own submission history as a measuring instrument: because each macro metric is a mean of a small number of rational per-question values, a controlled change to one prediction file often admits exactly one arithmetic explanation. Two of our predicted deltas came back exact to four decimals. We report the conventions this recovered, six heuristics it refuted, four bugs it exposed in our own verifiers, and the limitation that a large share of our final score reflects per-question auditing and leaderboard-feedback attribution rather than a system that would generalise.

1 Introduction

LitTraceQA asks a system to answer a research question by returning three things at once: the canonical identifiers of the paper or papers involved, coarse evidence locations inside those papers, and the answer in a requested format. Because these are scored separately, a system can be visibly right for the wrong reasons — a correct multiple-choice answer attached to the wrong paper, or the right paper with a locator pointing at the wrong page.

Our participation had an unusual shape and we state it plainly. We built a conventional pipeline — dense retrieval over the released metadata pool, cross-encoder reranking, an LLM selector, and a vision-capable reader for tables and figures — it reached an official 0.4563, improved to 0.5519 with selection and visual reading, and then stopped. Every subsequent gain, to 0.7649, came from work better described as *instrument reading* than as modelling: studying the scorer, recovering the annotators' conventions from the development split, and using controlled submissions to identify which individual predictions were wrong.

We believe the findings are more transferable than our architecture, so the paper is organised around them.

Contributions.

1. A decomposition of the official metric showing that two-thirds of the score is *provenance* rather than answering, and that the set-based evidence and table metrics make the cost of hedging exactly computable (§2).
2. Eight annotation conventions recovered from 55 development examples, each reported with the count it rests on, including two that transferred to test and produced our largest single jumps (§4).
3. *Score-guided attribution*: using a macro metric over a small test set to prove an individual prediction right or wrong, with two predicted deltas confirmed exactly (§5).
4. Six negative results and four bugs in our own instrumentation, reported so that others do not repeat them (§8).

5. An error taxonomy for grounded scientific QA in which the dominant failure is not hallucination but *the right paper and the wrong row* (§7).

2 The Metric Is Asymmetric

The released evaluator computes

$$\text{answer} = \tfrac{1}{3}\,(\text{mc} + \text{row\_f1} + \text{cell\_acc})$$

$$\text{overall} = \tfrac{1}{3}\,(\text{paper\_f1} + \text{evidence\_f1} + \text{answer})$$

so paper F1 and evidence F1 carry weight $1/3$ each, while multiple-choice accuracy, table row F1 and table cell accuracy carry $1/9$ each. This is the single most consequential fact about the task and it is easy to miss: an intuition that answer accuracy is the objective puts effort in the wrong place. On our first submission multiple-choice accuracy was already 0.68 while evidence F1 was 0.3587.

Evidence keys are coarse. Predictions and gold are compared as sets of tuples $(\textit{paper\_id}, \textit{source\_type}, \textit{page}, \textit{object\_id})$, where *object_id* is a normalised visible identifier for `table`, `figure`, `equation_algorithm` and `citation_context` and is *empty* for `text_span`. Three consequences shaped our work. (i) A `text_span` matches on paper and page alone, making it the cheapest correct guess when the type is uncertain. (ii) Citing the right page with the wrong type scores zero: "we found the answer" is not "we located it as the annotator did". (iii) Emitting N predictions against G gold items with C correct gives $F_1 = 2C/(G+N)$, so the cost of hedging is arithmetic rather than taste. We use (iii) throughout.

Cell accuracy ignores row precision. Row F1 is a set F1 over row-key tuples, so duplicate row keys silently collapse in the evaluator's dictionary. Cell accuracy iterates over *gold* rows only: cells in a predicted row whose key does not match gold are never examined, and unmatched gold rows count their cells as wrong. Cell accuracy is therefore bounded by row *recall* and indifferent to row precision — an extra candidate row can cost row F1 but can never cost cell accuracy. Including this term moves the break-even hit rate for adding a candidate row from about 60% to **26%**, which changed several of our decisions.

3 System

Retrieval and selection. The pool contains 27,487 papers with title, abstract, venue and PDF URL. We embed the question and title–abstract text, retrieve a candidate list, rerank with a cross-encoder, and pass the top candidates as a numbered list to an LLM selector that returns indices. Reranking moved paper F1 from 0.410 to 0.490 on a development family; the LLM selector with a cap of three papers per question was our best configuration.

We measured shortlist recall separately to locate the bottleneck: on a test-like development family, recall of the gold paper was 0.846 at rank 1 and 0.962 at rank 20, and *flat* out to rank 200. Widening the shortlist was therefore pointless and we did not pursue it.

Reading. Selected PDFs are read locally with PyMuPDF for page text, caption indices and page rasterisation. Answer generation is a per-question prompt carrying the question, the requested answer types, the supplied table schema, and page text or rendered page images. Table questions use a two-stage scheme in which row keys are extracted from the question before values are read, guarded by a check that every extracted key occurs in the question; without that guard the stage fabricated entities and cost 0.129 overall.

Audit loop. The component responsible for most of our score is a procedure rather than a model. Each candidate submission is checked question by question against the source PDFs: does the cited page contain the claim, does the cited object id appear in that page's captions, does every emitted paper mention an entity the question names, and does the chosen option match the row and column the question actually asks about. Adjudication was performed by an LLM agent reading page text and rasterised pages, with every decision recorded against a specific page of a specific PDF. This is a human-in-the-loop system in the sense that matters for interpreting our score, and we return to it in §10.

4 Recovered Annotation Conventions

The development split has 55 examples with gold papers, evidence and answers. As a validation set for tuning it is too small to separate configurations. As a corpus describing *how the annotators behaved*

Convention (counted on 55 dev questions)	support
Exactly one evidence key per gold paper	43/55
Gold table rows = gold paper count (multi-paper)	**8/8**
Question names a figure ⇒ gold has `figure`	**10/10**
Question names a table ⇒ gold has `table`	3/3
Asks an equation/loss ⇒ gold has `equation_algorithm`	4/7
Cited page carries a table caption ⇒ gold type is `table`	52/68
Gold pairs a `text_span` with a same-page object	1/64
Gold page is the *earliest* occurrence of the value	10/38

Table 1: Conventions recovered from the development split. Rows 2–3 transferred to test and produced our two largest jumps. The last two are why we did *not* ship a positional page-rewrite rule or a same-page `text_span` pad.

it was the most valuable resource in the task. Table 1 collects the conventions with their denominators, because several rest on small samples and a reader should see that rather than take them on trust.

One evidence key per paper. Distinct evidence keys versus gold papers gives $(1, 1)$ 24 times, $(4, 4)$ 19 times, plus $(3, 3)$ and $(9, 9)$. Gold evidence is provenance *per paper*, not an exhaustive list of supporting locations, and we sized our evidence sets accordingly.

Source type follows the question's wording. Conditioning gold's type on the question's vocabulary is strikingly predictive (Table 1). Acting on it produced our largest evidence gain: we had been emitting *no* `equation_algorithm` items at all, and adding them alongside `text_span` on six equation-form questions coincided with official evidence F1 moving from 0.6610 to 0.7121 (that submission also re-aimed two pages, so the delta is not attributable to the type change alone). Where gold's type is genuinely mixed, carrying both is the right arithmetic: against a single gold item, two predictions with one hit score $2/3$ where committing to one scores 0.4 in expectation at $P(\text{equation}) = 0.6$.

Gold table rows equal gold paper count. Every multi-paper table question in the development split has exactly one gold row per paper — 8 of 8, no exceptions. This does more than cap row F1. If gold has two rows, then gold's two keys *cannot* be two of our four quantity descriptors, because a two-row table cannot key its rows by four separate quantities. Three of our test predictions violated the constraint and were, by that argument, scoring zero.

Row keys reproduce printed labels. Where a question names entities, gold keys are those names lowercased, keeping qualifying parentheticals. Where it implies variants of one paper, gold uses the labels *printed in that paper's own table* — one development question keys its rows `w/ ground truth` and `w/ cube rcnn`. Checking every test row key against its source this way found four mismatches, the decisive one being a retrieval question whose gold keys carry the paper's `+Fake2` suffix rather than the bare dataset names we had used. Correcting it took that question from row F1 0 to 1.0 and cell accuracy 0 to 1.0 in a single step. We *added* the variants rather than substituting them, which is what guaranteed the eight cells behind those rows would be scored under either convention; a wrong bet would have zeroed them too.

Evidence pages are not the first mention. Gold's page histogram peaks at pages 6–7 (65 of 149 items) with only 5 of 149 on page 1, while our predictions put 40% of their items on pages 1–3. The obvious correction is nevertheless wrong: among gold items whose value appears on more than one page, gold is the earliest occurrence 10 times, in the middle 22 times, and the latest 6 times. There is no positional rule to learn, only a semantic one. We used the statistic to triage items for manual review and did not ship it as a rewrite.

5 Score-Guided Attribution

The task permits five evaluator submissions per task per day. Because each macro metric is a mean of a small number of rational per-question values, a submission that changes one thing often has exactly one arithmetic explanation, so the feedback identifies *which* predictions are wrong rather than merely whether the file improved.

Worked example. Table row F1 sat at exactly 0.340476 for five consecutive submissions. When two questions' row keys were changed and the macro moved by $+0.011111$ — that is $+0.2333$ over the 21-question sum — the shift decomposed uniquely as one question going from zero matched rows to one while its row count went from 2 to 4 ($2{\cdot}1/6 - 0 = +0.3333$) and a second question's added candidate missing ($2{\cdot}1/5 - 2{\cdot}1/4 = -0.1$). That established both that gold's `paper` row keys are short names rather than full metadata titles, and that one specific candidate string was wrong.

Reading exact fractions. `table_cell_accuracy_micro` is *cell_correct*/*cell_total* and *cell_total* is a property of gold. Read as an exact fraction it gives $23/87$ and later $31/87$: gold has exactly **87** comparable cells across the 21 table questions, and the $+8$ was one question's eight cells.

Predictive accuracy. Two predictions made before submitting came back exact. Dropping rows proven not to match was predicted at $+0.2952$ over the row sum and returned $+0.2952$; restoring three row sets was predicted at $+0.2500$ and returned $+0.2500$. A third prediction was wrong by -0.25, and because three of the changed questions had identical key sets before and after — so their contributions had to cancel — the shortfall forced the conclusion that a change we had reasoned was "free" had discarded a matching key. We reverted it and recovered the exact amount.

Zero-cost slots. The same arithmetic identifies predictions that cost nothing to change. If a question has $C = 0$ its row F1 is zero for any N, so replacing its keys is free. Row F1 was unchanged to six decimals after we added one row to each of two questions; adding a row can only leave F1 fixed when $C = 0$, and no other combination of their possible deltas sums to exactly zero. Both therefore score zero on rows — meaning `FVT` and `AP-Attack` do not match gold *despite being exactly what the question calls those things*.

Disclosure. This is adaptation to the held-out split. It is within the stated submission rules, and we report it because it materially explains our final number; scores obtained this way should not be read as an estimate of how the underlying system would perform on unseen questions.

6 Official Results

Table 2 reports official evaluator output for our scored submissions on `littraceqa-test` (71 questions). Every figure is as returned by the evaluator; no number in this paper is computed by us from private labels. We did not submit to the optional `littraceqa-test-extra` diagnostic track.

Where the score came from. Between our first and best submission, paper F1 rose by 0.338 and evidence F1 by 0.373; multiple-choice accuracy rose by 0.30 but contributes at one third the weight. Table row F1 and cell accuracy remain our weakest components at 0.4992 and 0.2976 and together hold the majority of our remaining headroom.

Automated verifiers. Five checks ran over every candidate file and are the reusable part of our tooling. (1) An *object-id verifier* regexes captions from each page and confirms the cited `table_id`/`figure_id` exists there; 48 of 49 of our ids passed and the single failure was a real error. (2) A *name-presence check*: does each emitted paper's text ever mention an entity the question names — a paper that never writes "ERASE" is not the ERASE paper. (3) The same check against all 27,487 titles, which finds papers never retrieved and therefore unreachable by any text-based check. (4) A *crowded-page check* flagging table items on pages carrying more than one same-type caption, where "the id exists on this page" does not imply the right table. (5) A *dead-run guard* failing any submission with all-null table rows; this caught two files silently produced after our API quota was exhausted mid-run, with no error anywhere in the trace.

7 Error Analysis

The dominant failure is the right paper and the wrong row. Of the answer errors we found and corrected, almost none are hallucinations. They are values genuinely printed in the cited paper, one row or one column from the one asked about: the unfinetuned base-model row instead of the fine-tuned one; `RPEt` where `RPEr` was asked; the `MC1` column where the question says `MC3`; inter-annotator agreement where the question says intra-annotator; a *Pearson* column read as a Kendall one; the *None* baseline row's accuracy reported as the proposed method's macro-F1; a cross-check-*enabled* row where the question says disabled. This is why three separate value-presence triages we built returned mostly false positives — every candidate value *is* in the paper — and why reading row and column headers is the only check that works.

Wrong papers hide behind plausible titles. Seven paper sets were wrong and every one was

Submission	paper F1	evid. F1	MC	row F1	cell acc	**overall**
initial pipeline	0.6324	0.3587	0.68	0.3221	0.1310	0.4563
+ LLM selector, visual table reading	0.7991	0.4737	0.78	0.2738	0.0952	**0.5519**
+ hand-authored table row keys	0.7991	0.4737	0.78	0.3405	0.0952	0.5602
+ answer audit against the PDFs	0.8554	0.5347	0.84	0.3405	0.1984	0.6166
+ audit continued	0.8789	0.5606	0.86	0.3405	0.2103	0.6366
+ seven paper-set corrections	0.9704	0.6610	0.94	0.3405	0.2103	0.7095
+ `equation_algorithm` evidence	0.9704	0.7121	0.96	0.3405	0.2103	0.7287
+ figure-counting answers	0.9704	0.7121	0.98	0.3516	0.2103	0.7322
+ short-name table row keys	0.9704	0.7121	0.98	0.4283	0.2262	0.7425
− rows proven not to match	0.9704	0.7121	0.98	0.4675	0.2262	0.7468
+ batched cell rewrites	0.9704	0.7192	0.98	0.4675	0.2500	0.7518
+ printed-label row keys	0.9704	0.7262	0.98	0.4873	0.2976	0.7617
− rows proven not to match (2)	0.9704	0.7262	0.98	0.5032	0.2976	0.7634
+ three row sets restored	0.9704	0.7262	0.98	0.5151	0.2976	0.7647
+ caption-semantics table swap	0.9704	0.7178	0.98	0.5151	0.2976	0.7619
− swap reverted, + two key unions	**0.9704**	**0.7319**	**0.98**	**0.4992**	**0.2976**	**0.7649**

Table 2: Official evaluator results on the LitTraceQA held-out test split, in submission order; each line names the change distinguishing it from the line above. Paper F1 and evidence F1 carry weight $1/3$ each in *overall*; MC, row F1 and cell accuracy carry $1/9$ each. Components are the evaluator's values rounded to four decimals, so recomputing *overall* from them can differ in the last digit. Row two is our best **fully automated** submission; every later row involves per-question intervention. The penultimate row *regressed*; we report it rather than only our best.

found by the presence checks rather than by re-running selection. Two required fetching a PDF that had never been retrieved, so no text-based check could have reached them; one was a question answered "all three papers" while citing a paper that was none of the three the options named.

Figures need rasterising. Four questions ask for counts existing only in a figure. Rendering those pages at 2× and reading them settled all four and corrected one answer: a paper states in prose that it classifies models into *eight* categories, and its pie-chart labels agree, where we had answered six.

Annotator-invented strings are not recoverable. Where a table's row-key column is a descriptor the annotator invented (`quantity`, `setting`, `attribute`, `metric`), we never matched it. Our record on guessing such strings is **0 for 13**: three candidate row keys in one submission (all missed), four successive forms of a single cell (all missed), two structural conventions on three questions (both missed), two convention unions (both missed), and the literal words the question uses for two entities (`FVT`, `AP-Attack`; both score zero). We report this as a limit of the approach rather than an unexplored lead, and it is the main reason our table components stalled.

8 Negative Results

We list heuristics that seemed reasonable, were measured, and were dropped. These are the most transferable part of the paper.

- **Padding paper sets.** From one submission's precision and recall alone, the gold set sizes on test are derivable as 1 or 2 (3 and 4 are arithmetically impossible), and our selector is size-calibrated: an uncalibrated selector cannot reach the observed precision even if its second pick were always gold. Padding singletons was refused.
- **Full-text indexing.** Content-defined multi-paper questions look like they need body text, but only 4% of the relevant development family is answerable from body text alone, and the shortlist was already saturated.
- **Re-aiming evidence at later pages.** Motivated by the page histogram; measured at $+0.013$ evidence F1 on development, below our shipping bar, and explained by the middle-position statistic.
- **Carrying both row-key spellings everywhere.** Measured at -0.020 row F1 on development, because there the bare keys are usually already right and the alternative never hit. We restricted hedging to keys we could *prove* wrong.
- **Converting `text_span` to `table` on caption-bearing pages.** The conditional is real ($52/68$), but all seven of our `text_span` items on such pages are verified prose, so there was nothing to convert.

- **Caption semantics as proof.** Our largest single regression. A question asks for a setting whose wording matches Table 2's caption almost verbatim ("expand the Cambrian-7M dataset with 2M pure text data") while Table 3 reads "using Cambrian 2.5M"; gold cites **Table 3**. Reverting returned exactly $+1.0$ of evidence sum, confirming it. Caption semantics is a useful heuristic and we had labelled it "confirmed", which was an error of epistemics rather than of arithmetic.

Four bugs in our own verifiers. Each briefly produced a conclusion we believed. (i) A multiple-choice triage stripped whitespace before matching numbers, which glues adjacent table cells together, so `16,825,000 5,800,000` became one token and both halves were reported absent; every "value not in this paper" verdict was worthless until whitespace was collapsed rather than removed. (ii) A caption regex matched in-text references such as "…in Table 3."; requiring the caption at line start raised the measured type conditional from 71.6% to 76.5%. (iii) A prediction-file diff compared a key that prediction files do not carry, so both sides read `None` and it reported "0 of 71 identical" when the true answer was 26. (iv) We reported a PDF as unreadable and concluded the pipeline never saw it; probing all 112 selected papers in isolated subprocesses gave *zero* unreadable, and the crash was resource exhaustion in our own script. We record these because an analysis pipeline is itself an experimental instrument, and ours was wrong four times in ways invisible until checked.

9 Setup and Disclosed Resources

Data. Released LitTraceQA metadata pool (27,487 papers), development split (55) and input-only test split (71) from the official Hugging Face dataset (CC BY-NC 4.0). PDFs were fetched from the publisher URLs in the released metadata. No additional annotated data was used and no synthetic training data was generated.

Models and APIs. Hosted `gemini-flash-lite-latest` and sibling Gemini models on the free tier for selection and reading; a locally served Qwen3 behind an OpenAI-compatible `llama.cpp` endpoint for text-only selection experiments, added because the hosted free-tier quota (500 requests/day) cannot score one development split twice; a cross-encoder reranker and a sentence embedding model for retrieval. The audit loop of §3 was performed by an LLM agent (Anthropic Claude) reading page text and rasterised pages. We fine-tuned nothing and trained no model.

Software and compute. Python; PyMuPDF for page text, caption indices and rasterisation; the organisers' released `evaluate.py` and `validate_submission.py` for local checks. All work ran on a single workstation with no GPU training.

Reproducibility. We release the pipeline, the five automated verifiers, the experiment scripts, and reports recording each measurement including the negative ones.

10 Limitations

The most important limitation is what our headline number represents. Our best *fully automated* run scores **0.5519** (row two of Table 2); the reported 0.7649 additionally includes per-question auditing against source PDFs and corrections attributed using leaderboard feedback. The audit is not a model that would run on new questions, and the attribution is adaptation to this particular split. A reader interested in what an automated system achieves on LitTraceQA should read the 0.5519 row. We report both prominently for that reason, and we include the submission that regressed rather than only our best.

Second, the conventions in §4 are counts over 55 development questions and several rest on small denominators (3/3, 8/8, 10/10). They were predictive on test, which is evidence, but they are not established at any conventional level of confidence.

Third, our weakest components are the table metrics, and our analysis explains why we could not fix them: reproducing a row key an annotator invented is not a reasoning problem with a findable answer. This interacts badly with the metric, because cell accuracy is gated on row-key recall, so a question whose key wording we cannot reproduce contributes zero to two of the five reported metrics no matter how correct the extracted values are. We regard this as a property of the task's table format at least as much as of our system, and we suggest a future edition might either publish the row-key vocabulary alongside the schema or score cells against gold rows independently of key matching.

Fourth, we did not submit to `littraceqa-test-extra`, so we cannot report how these findings transfer to its larger question pool.

11 Ethics

Papers were retrieved from publisher-provided URLs in the released metadata and used only for evidence extraction, within the dataset's CC BY-NC 4.0 terms. The purpose of a system like ours is to make claims about scientific literature checkable by pointing at a specific page of a specific paper, which we consider a beneficial direction. The corresponding risk is that a confidently formatted citation is more persuasive than an uncited claim even when wrong, and our own error analysis shows the dominant failure is a citation that is *almost* right — correct paper, adjacent row. Systems of this kind should surface the located span for human confirmation rather than presenting a grounded answer as settled.

12 Conclusion

Team OdeD's LitTraceQA submission reached an official 0.7649 from a 0.4563 start, and the useful part is where the gain came from: two-thirds of the metric is provenance rather than answering, the annotators' conventions are recoverable from 55 examples, and a macro metric over 71 questions is a precise enough instrument that a controlled submission can prove an individual prediction wrong. We also report what this process refuted, the four bugs it exposed in our own verifiers, and how much of our number reflects auditing rather than a system. For future participants we would compress the task into one sentence: locate the evidence the way the annotator located it, and check the row and column headers before believing a value you have found.

Acknowledgements

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References

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